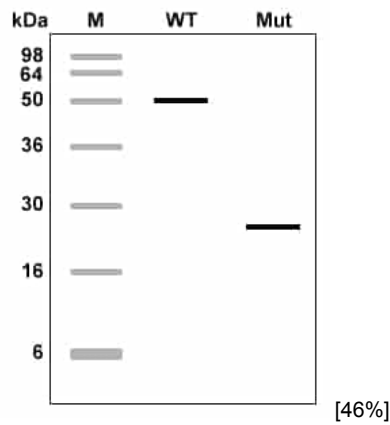
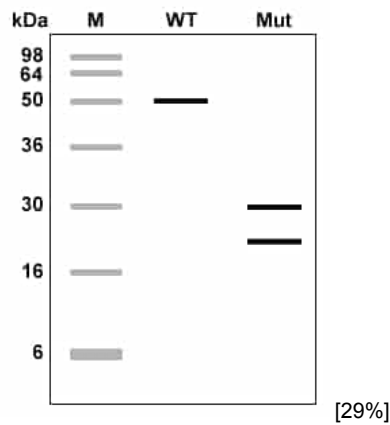


In an experiment, a single cysteine (C) in a wild-type (WT) protein was replaced with the amino acid serine (S) to create a mutant protein, and both proteins were compared on a nonreducing SDS-PAGE gel. Which of the gels below shows the expected result if the WT protein normally exists as a disulfide-linked homodimer?

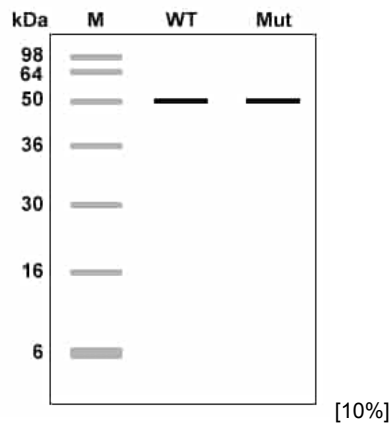
☒ A.



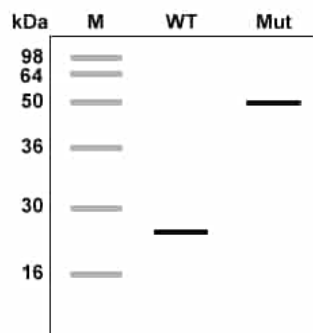
☐ B.



☐ C.



☐ D.



6

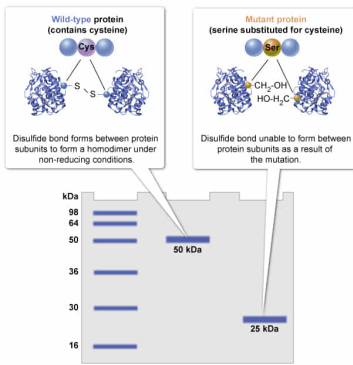
[13%]

Correct

45% answered correctly

Explanation:

Non-reducing SDS-PAGE reveals dimerization via disulfide bonds



Sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) is a technique that is used to separate **proteins** by **size** (ie, molecular weight). SDS is a detergent that unfolds and coats the protein with a uniform negative charge proportional to the protein's size. During electrophoresis, smaller proteins travel through the gel faster than larger proteins.

Treatment with SDS alone does not disrupt **disulfide bonds** within or between proteins. To distinguish the contributions of disulfide bonds to quaternary protein structure, SDS-PAGE gels can be run under **nonreducing** and **reducing** conditions. Under nonreducing conditions, disulfide bonds remain intact; under reducing conditions, a reducing agent (eg, β -mercaptoethanol) is added in addition to SDS, breaking disulfide bonds.

In this question, a WT protein exists as a disulfide-linked **homodimer**. An experimentally induced mutation results in the replacement of a **cysteine** with a **serine**, disrupting disulfide bond formation. The WT and mutant proteins were compared using nonreducing SDS-PAGE.

Because the **mutant protein lacks a disulfide bond** joining the two homodimers, and because nonreducing SDS-PAGE **does not disrupt the disulfide bond** in the **WT protein**, the **mutant protein** is expected to be **half the size** of the **WT protein**. Therefore, due to its **larger** size, the **WT protein** travels through the gel **more slowly** than the **mutant protein**. Only the diagram shown in Choice A demonstrates this result.

(Choice B) In this gel, the mutant protein shows two **different**-sized bands, the expected result when a protein exists as a heterodimer. However, because this individual protein exists as a **homodimer**, the two subunits are **identical** and would therefore run at the **same** size (ie, display only one band).

(Choice C) This gel shows the mutant and WT proteins running at the same size. A result such as this would be expected under reducing (*not* nonreducing) SDS-PAGE, because reducing conditions would break the disulfide bond in the WT protein.

(Choice D) This gel shows the WT protein as smaller than the mutant protein (ie, the WT protein travels farther than the mutant protein). However, if the mutation disrupts dimerization, the mutant protein should be **smaller** and travel **farther** than the WT protein (*not* vice versa).

Educational objective:

SDS-PAGE gels can be used to separate proteins based on their size. Under reducing conditions, disulfide bonds are broken whereas under nonreducing conditions, disulfide bonds remain intact.

Time spent: 0 sec

Subject:
Biology

QID:404086

System:
Molecular Biology